

GenAI

Hands On 2

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Basics of LangChain

LangChain is a tool that simplifies working with language models. Instead of setting up each model manually, you call a single function and choose the model as an option. This makes it easy to switch between different models without rewriting code.

Temperature controls how unpredictable the model's output is, with values from 0 to 1. A higher value produces varied responses, while a lower value makes outputs consistent. Running the same prompt at high temperature gives different results each time, but at zero temperature it produces identical answers.

A system message defines how the model should behave, such as tone or style, while a human message is the actual query. The system message acts like a role or personality for the model. The output parser simply cleans the result and returns plain text.

The workflow of an application follows input, prompt, model, parser, and output. These steps can be connected smoothly using a chaining operator instead of writing each step separately.

Prompt Design

A vague prompt gives unclear results because the model fills in missing details on its own. A well defined prompt includes context, purpose, and constraints, leading to clearer and more accurate responses.

In zero shot learning, the model answers without any examples. In few shot learning, examples are included so the model can follow patterns and improve accuracy. When there are too many examples, a selection method is used to choose the most relevant ones.

Reasoning Methods

Step by step thinking improves accuracy because the model breaks problems into smaller parts. Smaller models are useful for testing because their mistakes are easier to observe

and fix. Direct questions may lead to errors, while guiding the model to think step by step improves results.

Reasoning happens through intermediate tokens that help the model arrive at better answers. Some methods explore multiple solution paths before choosing the best one, while others combine different ideas into a final response.

RAG and Vector Concepts

Retrieval augmented generation improves responses by adding external information before generating answers, reducing errors.

Text is converted into numerical vectors so machines can understand meaning. Similar content produces similar vectors in a multi dimensional space. Similarity between vectors is measured using angles, where smaller angles mean closer meaning.

The RAG process involves retrieving relevant data, combining it with the query, and generating a response. Vector databases store these representations and allow quick searching.

Model knowledge comes from training, while external data acts as an additional source to keep answers accurate and updated. Efficient indexing methods help search large datasets quickly without comparing every entry.